

Institute of Actuaries of India

Subject SP4 – Pensions & Other Benefits

July 2022 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

Risks of the plan from the perspective of the company

- Longevity risk – since it is a DB Plan, the employer bears the risk of retirees' life expectancy for any retiree that elects an annuity. Since the vast majority of participants may opt for a lump sum at termination/retirement, longevity risk may not be a large risk for such a plan.
- Interest rate risk – risk of shifts in the interest rates used to measure the liability and that it will not parallel the shift in market value of assets. Hybrid plans have a short liability duration.
- Investment risk – risk of return on plan assets. Hybrid plans over the long-term attempt to earn more on the plan assets than the interest credit rate provided to participants. Investment risk is increased if the plan offers a minimum interest credit.
- Litigation risk is slightly higher than traditional DB plans since hybrid plans are a newer plan design.
- Other DB plan risks, such as plan funding risk, compliance/administration risk
- Anti-selection risk for those that take a lump sum distribution versus an annuity

[Max 5]

ii)

Ways a company could manage the risks of this pension plan:

- The investment risk would be reduced if a smaller interest credit rate was offered.
 - A cumulative minimum rate is less risky than an annual minimum interest crediting rate.
 - The plan can be designed to credit the actual return on plan assets, which would eliminate investment risk (other than the floor of zero and for retirees).
- Fund the plan adequately to avoid cash flow surprises, being careful not to overfund the plan too much.
- Manage liquidity risks – participants are likely to take a lump sum so allocations to illiquid assets should be moderated.
- Being aware of untested designs or practices which could expose the plan sponsor to lawsuits.
- Invest in assets that have shorter durations to match the shorter liability duration, lowering interest rate risk.

[Max 5]

iii)

Defined Contribution Plan	
Employee: ○ Opportunity to earn excess investment return (advantage) ○ Possibly loan and withdrawal provisions (advantage)	Employer: ○ Cash cost is more predictable (advantage) ○ No accrual accounting required (advantage) ○ Risk (investment and longevity) shifted to the employee (advantage)

More risk (investment and longevity) (disadvantage)	- More understandable by employee (advantage) - Cannot use the plan as a workforce management tool (early retirement subsidies, windows, etc) (disadvantage)
Variable Defined Benefit Plan	
Employee: - Change of earning excess investment earnings (advantage) - Can produce smaller benefit if investments underperform (disadvantage) - Benefit may change in retirement (disadvantage)	Employer: - Investment risk shifted to employee (advantage) - Longevity risk remains with employer (disadvantage) - Uncommon design, which may entail legal risk (disadvantage)
One time salary increase	
Employee: - Has access to cash for whatever purpose they want to use it for (advantage) - Does not provide retirement income (disadvantage) - Immediate taxation of amount (disadvantage)	Employer: - Predictable cost (advantage) - No risk (longevity or investment (advantage) - Cannot be used for workforce management (disadvantage) - More valued by younger than older employees (advantage and disadvantage, depending on employer objectives)

[Max 10]

[20 Marks]

Solution 2:**i) Key assumptions**

- Discount rate
- Medical inflation rate
- Claims costs
- Premium costs
- Mortality – post retirement
- Mortality – in-service
- Attrition rates
- Disability rates
- Percentage married
- Spouse age difference

[Max 4]

ii)

Reasons why prevalence of post-retirement healthcare plans may be reducing and why plan sponsors may be looking to reduce such benefits

- The benefits are valuable for a minority of active employees; employees start valuing the benefits only after they start considering retirement (i.e., only appreciated by employees when they start thinking about retirement).
- Historical employment relationship that focused on long-term career may be diminishing.
- As workers assert their independence and their lack of loyalty to the employer, the sense of providing retiree healthcare as a social responsibility is quickly evaporating.
- As more of the competitors eliminate retiree health care coverage, it makes it easier to drop or reduce the benefit.
- The cash cost of retiree health care is increasing faster than any other cash item for most companies.
- The liabilities and ongoing accruals reported under any financial statements in accordance with generally accepted accounting principles may be increasing rapidly.
- There is continuing steep increase in plan costs due to participant count increases, anticipated increases from price inflation and additional utilization of medical goods and services.
- Group payments increase for a closed group many years after the count has begun decreasing, possible beyond 50 years, due to aging affect and mostly due to compounding effect of health care trend.
- Health care benefits may not be tax efficient for employers
- Unions may have shown willingness to negotiate with plan sponsors to eliminate or reduce health care benefits.

[Max 6]

iii)

Option 1: Make large lump sum contribution to the approved DB pension plan

Active employees	
Advantages: • The active employees may be happy with contribution to the underfunded plan because it improves benefit security of accrued benefits. • It reduces probability of involuntary plan termination or plan freeze of future accruals. • It reduces probability of benefit restrictions - restrictions on lumpsum payments or ceasing future benefit accruals.	Disadvantages: • Active employees may prefer a one-time cash bonus. • Active employees may prefer a salary increase.
Shareholders	
Advantages: • Large lump-sum contribution to the approved plan could result in improved funded status, reduced future contribution levels and pension expense volatility; potentially increasing shareholder equity.	Disadvantages: • Shareholders might misconstrue contribution to the plan as an imprudent investment by the employer; it may result in a decline in shareholders' equity.
Company	
Advantages: • If contribution does not exceed the maximum deductible contribution limit recommended by the actuary, then the contribution to the plan	Disadvantages: • The employer may have been able to earn a higher rate of return (net of taxes) if invested large

may be tax effective because contributions to qualified plans are tax deductible. • Large lump-sum contribution to the approved plan may improve funded status and reduce future contribution volatility. • Large contribution may improve funded status of plan, reduce volatility on balance sheet; improve credit rating and increase shareholders equity. • It reduces probability of benefit restrictions.	contribution in company vs. in trust fund of pension plan, thereby increasing the value of the company and improving credit rating of the company. • The employer could use the extra cash to pay off creditors.
--	---

Option 2: Pay a one-time payroll bonus that would be immediately taxable to the employees

Active employees	
Advantages: • Beneficial to employees in the lower income tax bracket as the one-time lump-sum payment may rise standard-of-living (immediately). • May boost employees' morale and result in increased productivity.	Disadvantages: • May not be tax advantageous to active employees in the higher income tax bracket. • Some employees may prefer a salary increase rather than a onetime bonus.
Shareholders	
Advantages: • None	Disadvantages: • Payment of bonus maybe misconstrued by shareholders as an imprudent investment by the employer; may result in a decline in shareholders' equity.
Company	
Advantages: • The employer may see the one-time bonus disbursement as a reward to employees and a way to help boost morale and increase employees' productivity.	Disadvantages: • May not be tax efficient for the company • The employer could use the extra cash to pay off creditors.

Option 3: Fund the PRMB plan

Active employees	
Advantages: • Active members entitled to benefits from this plan will be happy because funding the benefits with a contribution will ensure that the benefits promised will be paid when due.	Disadvantages: • Depending on funding vehicle used to fund the plan, active employees might be subject to immediate taxation of their benefits because of the contributions.
Shareholders	
Advantages:	Disadvantages:

Current shareholders are paying for current benefits.	Funding the plan may be misconstrued by shareholders as an imprudent investment by the employer may result in a decline in shareholders' equity.
Company	
Advantages: - Large lump-sum contribution to the plan could result in lower pension expense; reduce balance sheet volatility and improve shareholders equity. - Large lump-sum contribution to the plan could improve perception of the value of the benefit and may result in improved morale and productivity.	Disadvantages: - It is less tax efficient than if employer were to contribute to the approved pension plan instead - Depending on the tax status of the company, funding the plan may not be tax efficient - The employer may have been able to earn a higher rate of return (net of taxes) if invested large contribution in company vs. in trust fund, thereby increasing the value of the company and improving credit rating of the company. - The employer could use the extra cash to pay off creditors instead of funding the plan.

[Max 20]

[30 Marks]

Solution 3:**i)**

The liabilities are extremely long term. The actual timing of the payments is unknown. Further the in-service liabilities increase as salaries increase and the pensions in payment are likely to also be increasing to some degree in line with inflation. The amounts of the future liabilities are therefore unknown.

Some liabilities may crystallise on discretionary events — for example a highly paid, long serving member leaving service or electing to take early retirement.

The total contributions may be volatile. If members' contributions are fixed then the company's contribution will be geared and thus more volatile.

While the returns from fixed interest stocks are known, there may be a default risk. The returns from equity and property investment are unknown, and assumptions will need to be made.

[Max 6]

ii)

Due to the uncertainties in amount and timing of both asset proceeds and liability outgo, full matching of cash flows will not be possible. One reason is that there may not be assets available of sufficiently long duration.

Over the long-term, shares are generally considered to be the most appropriate asset to meet long-term real liabilities. Indexed linked bonds would also be appropriate. For fixed liabilities such as pensions in payment, a substantial proportion of fixed or index linked bonds would be appropriate.

Most asset types have uncertain returns. If the changes are not managed over time, then there would be large trading costs associated with one-off changes. The costs of trading to adjust the portfolio frequently to achieve full matching may be prohibitive.

Given infinite resources, it is possible to try to meet outgoes by buying excessive amounts of securities. In practice, the matching portfolio is the portfolio which costs the least and which still provides the required certainty of meeting the liabilities. The scheme may have to alter its investment guidelines to allow this.

[Max 6]

iii)

- It is likely that the regulators would want to ensure that pension schemes are in a position to fulfil their obligations to members i.e. meet liabilities, or remain solvent. [1]
- The scheme may have been deemed to be insolvent by the regulator and hence forced to close. [1]
- To that end, they will require schemes to hold sufficient assets and make sufficient contributions. [1]
- This could lead to volatile contributions causing uncertainty. [1]
- They will probably prescribe the bases to be used to analyse such sufficiency. [1]
- In order to ensure, as far as possible, that assets comfortably exceed liabilities bases will tend to be conservative. [1]
- In particular with regard to the key assumptions of mortality (low in retirement) and investment returns (risk free returns). [2]
- This will mean that the employer will need to tie up a significant amount of excess capital in the scheme. [1]
- If the scheme turns out to be underfunded on the regulatory basis, the employer may be required to make good any shortfall in a short period of time. [1]
- Or the amount required may be large. [1]
- Similarly, accounting regulations may distort profit and losses (or balance sheet) due to the treatment of pension costs. [1]
- Putting a strain on resources needed elsewhere. [2]
- Regulation must be paid for – hence contributions to a compensation or protection scheme may be required. [1]
- Regulations may dictate the level and type of benefits to be provided by the scheme or the level of contributions. [2]
- In particular in reference to leaving service and to increases in deferment and payment. [1]
- Provision of these benefits may be costly e.g. if an inflation link is required. [1]
- Regulation may restrict the investments of the scheme. [1]
- It is likely that the scheme will be forced to invest in low-risk assets e.g. government bonds, which will reduce investment returns. [1]
- A requirement to match assets and liabilities (or hold mismatching reserves) will have a similar effect on investment returns. [1]
- Regulations are likely to be complex and so costly and time consuming to comply with. [1]
- Possible regulatory changes will introduce uncertainty – changes invariably imply extra costs or complications. [1]

[Max 10]

iv)

The options that might exist for the provision of the outstanding benefit payments from a discontinued scheme are:

1. gradual removal of liabilities through temporary continuation
2. transfer the liabilities to another pension scheme with the same sponsor
3. transfer the funds to the beneficiary to extinguish the liability
4. transfer the funds to an insurer to invest and provide a benefit
5. transfer the liabilities to an insurer to guarantee the benefit
6. transfer the liabilities to a central discontinuance fund.

[Max 3]

v)

- They should have considered recent expense levels. [1]
- Relative to any relevant expectations. [1]
- To appropriately earmark the expenses, the expenses should be broken down and categorised into significant distinct groups. [2]
- For example; property related, staff costs, investment, technical costs (systems maintenance say), professional services etc. [2]
- It may also be useful to subdivide these groups if large distinct items exist e.g. bills, rent, taxes etc. for property. [1]
- A split between fixed and variable expenses may be useful. [1]
- Expenses may be linked to membership and so a projection of membership movements over a relevant timeframe (life of scheme) will help. [2]
- Large one-off items or abnormal events should be stripped out to give an idea of typical ongoing expenses. For example, allow separately for closure costs. [2]
- To this end, analysing against past expected expenses may help with identification. [1]
- Likewise, from past data, it may be possible to spot trends and so help earmarking future expenses. [1]
- When earmarking based on recent experience, it will be necessary to allow for future inflation of expenses. [2]
- It will be important to allow for the correct inflation measure e.g. salary or price or “tax” linked. [1]
- Some allowance should be made for future one-off expenses e.g. repairs, redundancy costs as staff numbers decline etc. Any planned projects etc. should be covered. [2]
- A further contingency margin to cover unexpected issues e.g. higher inflation should also be included. [1]
- However, past expenses may not be a good guide especially as they should decline as the scheme winds down. So more detailed projections will be needed to allow for predictable developments. [3]
- Even then, the past expenses used as a basis may be inaccurate e.g. if some subsidy from the employer was present – shared staff etc. [1]
- It is possible that controls were lax in the past i.e. actual expenses should in reality be a lot lower. [1]
- Advice should be sought from other administrators or consultants to determine what expenses should be for a scheme like this. [2]

- Even then, resources might be so stretched that the Trustees require further cuts so as to reduce expenses even further – i.e. an overall resources constraint. Alternatively, a significant surplus may imply a laxer approach [2]
- Availability of outsourcing could give an idea of costs (and would be the cost if done). [1]

[Max 12]

vi) Why overspending?

- A significant factor could be issues surrounding the closure of the scheme. [1]
- The process may have taken a lot longer and been more complex than assumed when earmarking for the expenses involved. [1]
- For example, significant research and consultation may have been needed in developing new investment strategies and tactics. [1]
- Buying and selling individual stocks may have costed more i.e. rearranging the portfolio. [1]
- Likewise, more legal, actuarial or accounting/auditing work may have been needed. [1]
- For example, regulatory issues or communicating with and determining amended benefits for members may have been messier than expected. [1]
- It is possible that expenses that should have been met by the employer have been charged to the scheme – e.g. advice to them explicitly. [1]
- Ongoing investment expenses could exceed the earmarked expenses towards it if managers are trading too much and so incurring dealing costs. [2]
- Or if managers are paid by performance and performance has been good. [1]
- One-off unforeseen jobs e.g. changes in legislation could have had an impact on professional fees. [2]
- Membership movements not being as expected e.g. lots of transfers, deaths or options exercised could mean higher investment, actuarial or legal costs – must link this to these costs not just general expenses. [1]
- There could be an element of overcharging e.g. the professional advisers were in a stronger bargaining position if the scheme had not researched properly
- what fees should be or unnecessary work done. [2]
- Likewise, the “A” team might have been used e.g. expensive consultants and expensive staff within each consultancy. [1]
- Ongoing work should be routine and hence does not really need top-flight advice. [1]
- Market forces of supply and demand may mean that inflation of professional fees has been higher than expected. [1]

[Max 8]

vii) Transferring funds to beneficiaries

The issues for the trustees:

- Security of the scheme – the ability of the scheme to provide benefits and in the given situation it appears safe to provide the benefits in hand. The impact on the security of the scheme for the remaining members or will members receive reduced transfer values? Are more funds available if there is a deficit based on the recent overspends? [1]
- Use of the funds – how will members use the funds? Will the funds be squandered, or will there be any restrictions on the use of the funds? [½]
- Fairness – will this option result in all categories of members being treated equitably. Are there any winners or losers against other possible options? The other possible options may be ruled out as there may be deficit in the scheme given the overspend during last three years [½]

- Investment policy – assets will need to be realised and this could result in uncertainty where there are volatile assets and potentially high dealing costs. Is it a good time to realise assets in the market as the economy overall is not good? [½]
- Risk – DB risks will be removed from the scheme and the risks will be transferred to the beneficiaries for those who take up this option. [½]
- Advice – is there a need to provide advice to members as to whether to take this option and, if they do, subsequently what to do with the funds? [½]
- Legislation, administration and communication considerations. [½]

[Max 5]**[50 Marks]**
